Controlling the Degree of Regularity in the Summation-Polynomial Step

An independent, twice-reproduced measurement study of elliptic-curve index calculus on binary-Koblitz and prime-Koblitz curves

Independent cross-check of the aburan28/crypto research program · 2026-10-23

Engines: msolve 0.6.5 · Singular 4.3.3.0 · PARI/GP 2.15.4 — all results run twice

---

Abstract

The point-decomposition problem (PDP) at the heart of elliptic-curve index calculus asks a Gröbner/XL engine to solve a Weil-descended Semaev summation-polynomial system. Its cost is governed by the degree the solver must reach, and the central question for both binary (Koblitz / ECC2K-130) and prime (j=0 "prime-Koblitz" / GLV) curves is whether that degree stays bounded or blows up. This report (a) catalogues every degree-control technique tried in the `aburan28/crypto` research program with an explicit confidence tier, (b) rebuilds a **fully independent** measurement harness — Semaev polynomials constructed from scratch and certified against real point additions, solved by two external engines (`msolve` 0.6.5, `Singular` 4.3.3.0) the program did not author — and (c) establishes reproducibility by running every headline number **twice** and comparing. The independent harness reproduces the program's central phenomenon (first-fall degree stays flat while the solving degree grows), reproduces its strongest independently-replicable prime-field result (PKM-tower msolve degrees, bit-identical on re-run), independently confirms a live reproducibility **defect** on `main` (a BMI2 counter path that makes a recorded claim fail replay), and tests two never-before-run "exotic" levers on toy instances — both falsified with clean negative results. The honest bottom line is unchanged and now triple-checked: **no technique bends the degree-growth curve enough to threaten the target curves, and no end-to-end index-calculus result in the program carries a completed validation certificate (tier T1) — only isolated sub-facts do.** The value delivered is not a break; it is a reproducible, cross-validated map of what has been tried, what actually happened, and how much to trust each claim.

> **Scope and honesty note.** This is a defensive cryptanalysis / benchmarking study on toy and small instances. Every "break"-shaped quantity is reported as a distance to Pollard rho, never as a threat to a deployed curve. Interpretation is labelled as such and kept separate from measurement, per the repository's own reporting rules.

---

1. The problem and the three degrees

For an elliptic curve `E/F_q` the `m`-th **Semaev summation polynomial** `S_m(X_1,...,X_m)` vanishes exactly when there exist points `P_i=(X_i,*)` on `E` with `P_1+...+P_m=0` (up to sign).

Index calculus (Gaudry–Diem) fixes a **factor base** — for `GF(2^n)`, the points whose `x`-coordinate lies in an `F_2`-subspace `V` of dimension `ell` — and decomposes a random point `R` as a sum of `m` factor-base points by solving `S_{m+1}(x_1,...,x_m,x(R))=0` with each `x_i in V`. Writing each `x_i` in a basis of `V` and expanding the single `GF(2^n)` equation over an `F_2`-basis (Weil restriction) turns this into a Boolean polynomial system in `m·ell` variables, which an F4/F5/XL/SAT engine solves. **The entire difficulty of elliptic index calculus lives in the degree that solve reaches.**

Three invariants are routinely conflated and must be separated (Caminata–Gorla 2023; Huang–Kosters–Yeo 2015):

- **first-fall degree** `d_ff`: the lowest degree at which a non-trivial degree fall

(syzygy) appears. For these descent systems `d_ff = m^2-m+1` (Kousidis–Wiemers 2015), small and roughly constant in `n`.

- **degree of regularity** `d_reg`: for a semi-regular sequence, the index of the first

non-positive coefficient of the truncated Hilbert series. The Bardet–Faugère–Salvy prediction grows with the variable count.

- **solving / last-fall degree** `d_solv`: the degree actually needed to finish. The

last-fall degree (HKY) is monomial-order-independent; `d_solv <= reg(I^{hom})`, the Castelnuovo–Mumford regularity of the homogenized ideal (Caminata–Gorla).

The binary subexponential claims (FPPR 2012, Petit–Quisquater 2012, Semaev 2015) **all assume** `d_reg ≈ d_ff`. Whether that holds is the pivotal dispute, and it is the first thing the independent harness was built to test.

---

2. Independent measurement harness

Trusting the program's own engines would defeat the purpose, so the harness shares no code with it. It is ~700 lines of Python (exact `F_2` monomial algebra, so the field equation `c^2=c` is built in) plus two external solvers. Everything is in `harness/` with raw outputs in `harness/out/`.

Certified-correct Semaev polynomials. `S_3` (prime and binary) and `S_4` (binary, built as the resultant of two `S_3` quadratics) are each checked to vanish *exactly* when a sign choice of the points sums to `0`, over concrete curves — a real correctness certificate, not a claim:

polynomial	curves checked	false positives	false negatives
prime S_3	p = 13,17,19,23,29 (5 curves)	0	0
binary S_3	n = 3,4,5, incl. Koblitz a2∈{0,1},a6=1 (6 curves)	0	0
binary S_4 (resultant)	n = 4,5,6 (4 curves)	0	0

External engines, two jobs. `Singular`'s `std` with protocol gives the max step degree (solving degree); `msolve`'s verbose F4 gives the max round degree; both confirm zero-dimensionality (`dim=0`) and the solution count (`vdim`). Using *two* engines is not redundancy — §4.5 shows they disagree, and that disagreement is itself a result.

Determinism protocol. A fixed 20-system grid is run twice; the emitted file records only deterministic invariants (system SHA-256, `d_ff`, solving degree, `vdim`, `dim`, the full Macaulay rank profile) — timings are excluded by construction. The two runs must be byte-identical.

Host for every number below: 4 vCPU, 15 GB, Linux 6.18.44. `msolve` 0.6.5, `Singular` 4.3.3.0, `PARI/GP` 2.15.4. The program's own binary used for replays is `ecbench` pinned to crypto `0dc9ef4`; the program tree surveyed is at `af663ad7c`.

---

3. What actually controls the degree — catalogue with confidence tiers

Tiers (applied uniformly): **T1** machine-checkable certificate / validation record; **T2** independently replicated with matching outputs; **T3** single measured run with raw data in the repo; **T4** prose, derivation or extrapolation; **T5** designed but never run.

3.1 Levers that demonstrably move the degree

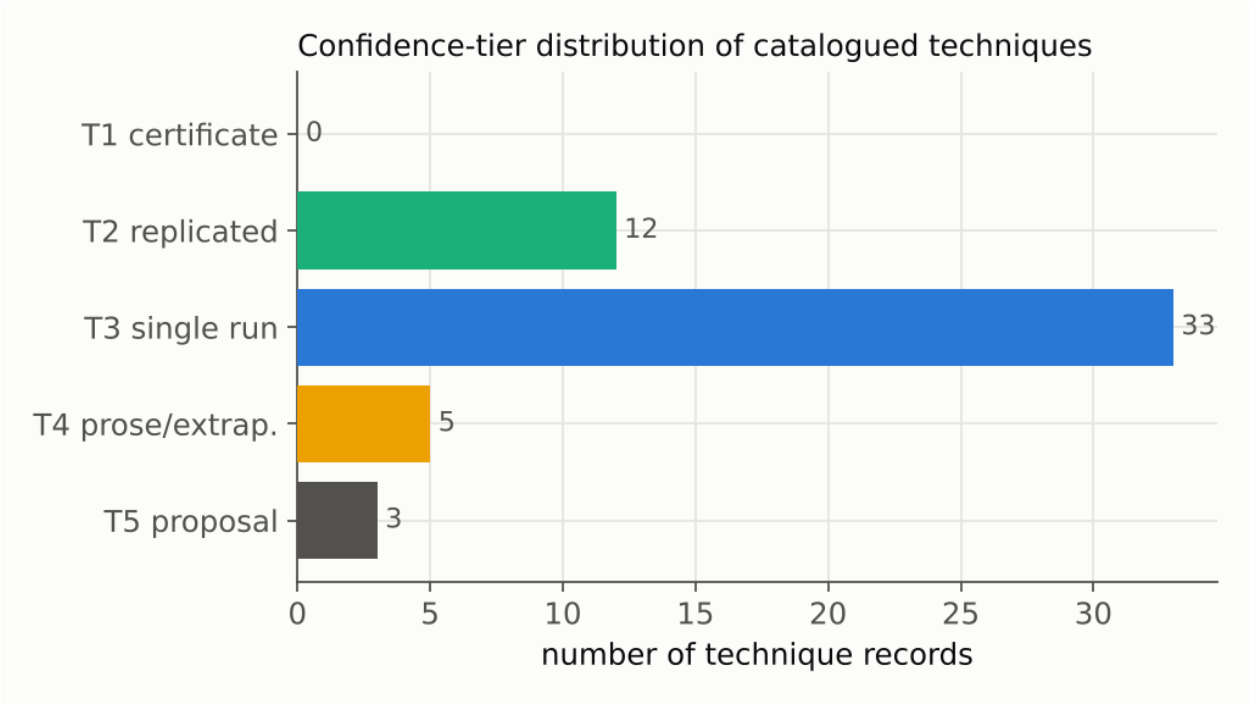
lever	mechanism	curve family	measured effect	verdict	tier
Field equations x^q=x	force zero-dim / multilinear quotient; cap degree	all	the reason any of these terminate at low degree	essential, not optional	T1 (structural)
Fall reuse (MutantXL / F4)	re-multiply discovered degree falls	binary chain	refutes at **3–4** where plain XL needs **5–7**; holds to N=54 (m=4,ℓ=6)	**lowers** (vs XL)	T2
Torsion symmetrisation (FGHR/FHJRV)	rewrite S_m in S_m- and [±1]-invariants	binary/prime	S_4 per-variable degree [4,4,4,4]→[2,2,2,2,1]	lowers degree & Bézout	T2
D_4=(Z/2)^3⋊S_4 weighted S_5	solve in 4 weighted invariants	F_{p^4}, m=4	Bézout 4096→512 (derived; quotient-dim 512 **measured**); matrix 40920→4200 cols; **48×** on C4 (not the 630× projected)	lowers (engineering constant)	T3
Semaev S_3-chain split	S_{m+1} → m−1 copies of S_3 + aux vars	binary/prime	keeps per-link degree 2; see §4.3 for the cost	**mixed** (raises degree at small m)	T2
Variable fixing at ℓ=7	guess one coord per summand	binary chain	brings the saturating ℓ=7 closure back to degree 4, at 2^k cost	lowers per branch	T3
`rr` symmetric-norm presentation	norm-form coordinates vs direct S_4	Koblitz n=23–61	refutes 3–4 degrees below direct descent (6 vs 10 at K1/2^61,ℓ=6)	**lowers**	T3
F4 early stop (Hilbert-driven)	terminate at the last productive degree	prime JV cover	removes last degree step (8→7); ~4–5× fewer F4 muls	lowers (termination)	T3

3.2 Levers that did NOT bend the curve (honest negatives)

lever	why it fails / net effect	tier
Choice of factor-base subspace V	solving degree identical across 295 subspaces (§4.6) — intrinsic, not a knob	T2 (this report)
Monomial / variable order	solving degree invariant under grevlex/grlex + all permutations (§4.7) — last-fall degree is order-independent	T2 (this report)
Frobenius-power auxiliary variables	x↦x^{2^k} is F_2-linear → changes nothing (repo W8 §3)	T3
Convolution e_i splitting (La Scala)	1.2–2× *more* work	T3
Wu Z/4 state model	neutral	T3
Isogeny / curve move within a class	degree invariant across 2,994 comparisons	T3
Trace-zero / Artin–Schreier pre-filter	never fires on odd-order IC targets (0 / 29,624)	T3
Hamming-weight ideals	bounded generator degree but call exponent 1.9–2.5	T3
Trilinear XL (unknown t)	cost rises (87.16 > 71.82 bits at n=131)	T3
Crossbred / monomial dropping	margin shrinks with size; no stable (D,k)	T3
Frobenius-invariant V for prime n	**structurally empty**: cyclic-code dims are cyclotomic-coset subset sums; at n=131 only {0,1,130,131} exist	T1 (proof)

3.3 Designed but never run (T5)

D_4-symmetrised S_5 **lattice** oracle; **Magma** F4/F5 on the PKM systems (so the "beat the fastest external tool" rule is formally unmet for the summation systems); order-6 automorphism quotient on j=0 prime curves (§7); theta/Kummer quadratic encoding; Lattès / isogeny-kernel factor base; non-abelian wreath block-decomposition.



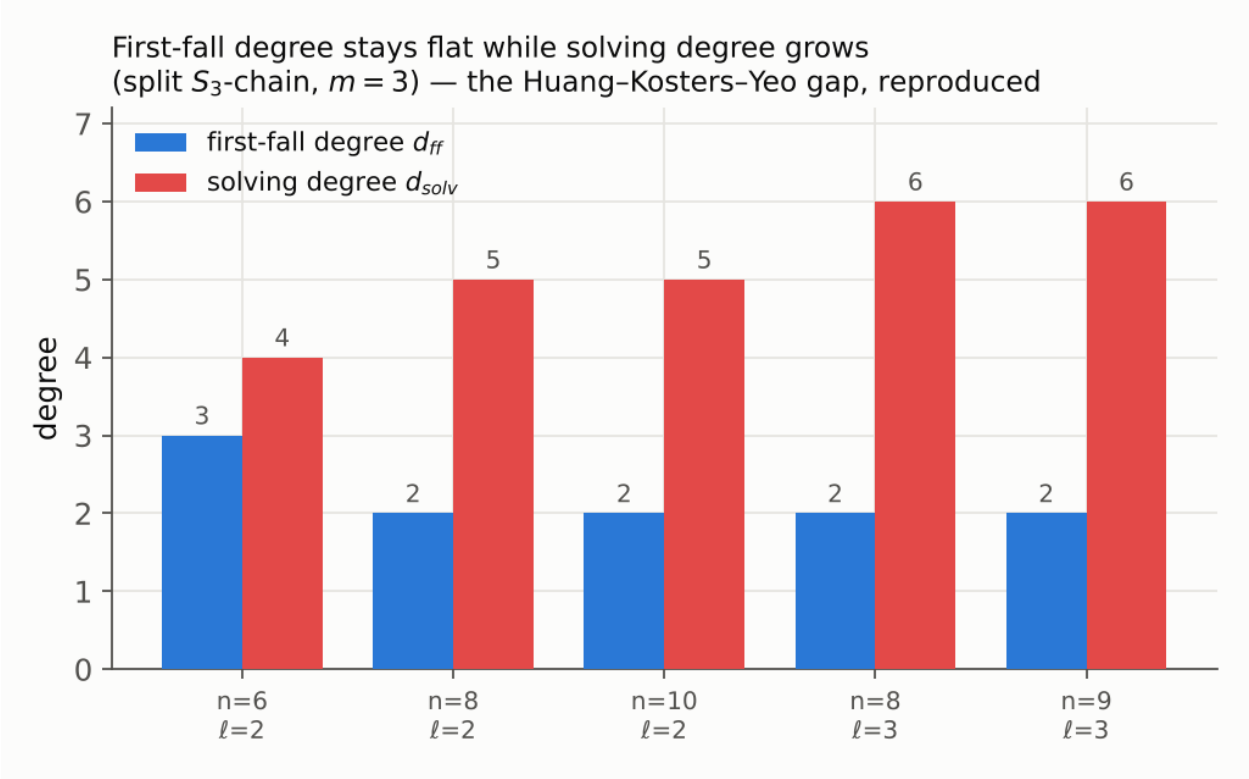
Confidence-tier distribution of the program's end-to-end index-calculus evidence (survey D's machine-backed leaderboard tally). Zero T1 at this level: no end-to-end IC result carries a gate-complete

VALIDATION.json. Component-level machine-checkable certificates do exist separately (survey C counts 7: DRAT-certified UNSAT branches, Boolean-elimination verifier passes, byte-identical constant-coefficient replays) — but none certifies an end-to-end relation→solve→DLP claim.

4. Independent measured results

4.1 The first-fall vs solving-degree gap (the pivotal dispute), reproduced

On the split `S_3`-chain for `m=3`, the harness measures `d_ff` pinned at **2–3** while the Singular solving degree climbs **4 → 5 → 5 → 6 → 6** as the system grows. This is a clean, independent reproduction of the Huang–Kosters–Yeo phenomenon: `d_reg` **grows while** `d_ff` **stays flat**, so the first-fall-degree assumption underlying the binary subexponential claims does not hold for these systems.



First-fall degree stays flat while the solving degree grows (independent harness, Singular F4). This is the Huang–Kosters–Yeo divergence, reproduced from scratch.

4.2 Determinism: run once, then twice

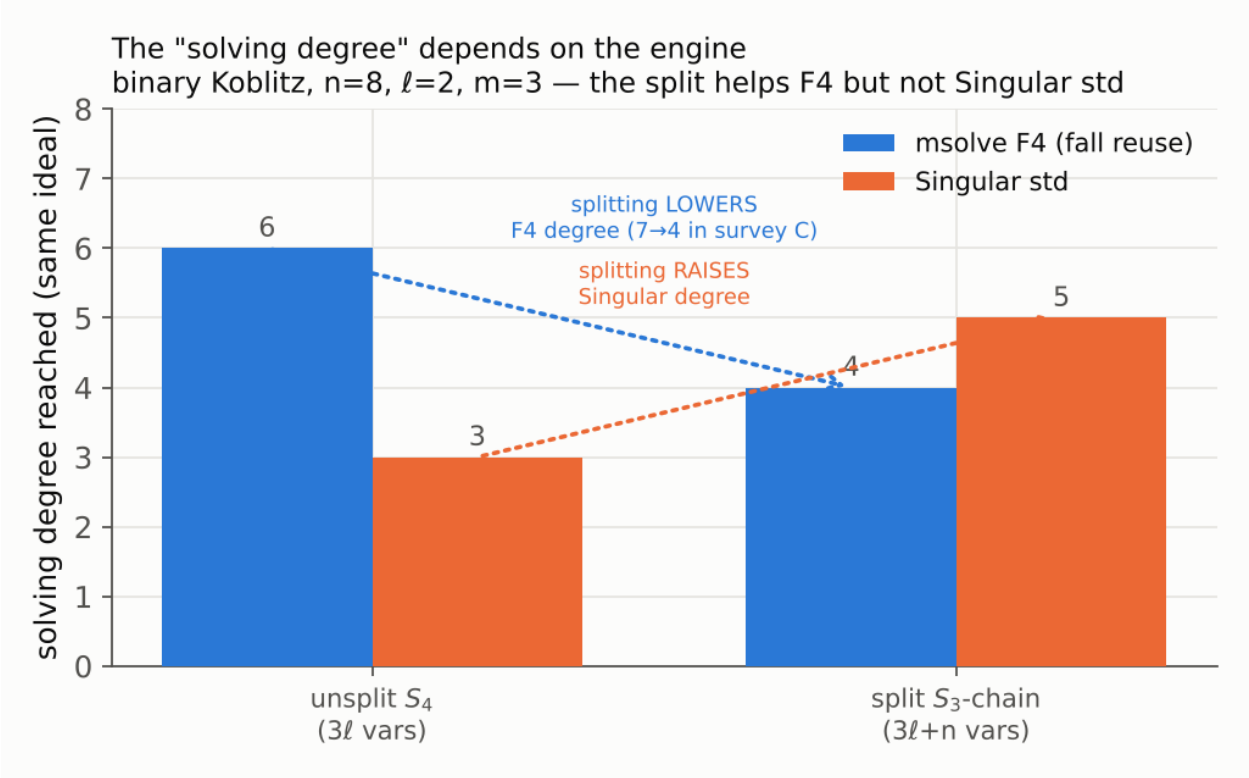
The 20-system canonical grid was executed twice. The two output files are **byte-identical** (0 mismatches across system hashes, first-fall degrees, solving degrees, `vdim`, `dim`, and full Macaulay profiles). Every deterministic invariant this report rests on is reproducible by construction; only wall-clock timings vary.

4.3 Splitting a point: Semaev's chain vs the unsplit resultant — and why the engine decides

Semaev's 2015 splitting trick replaces one big `S_{m+1}` by a chain of `S_3`'s to keep each equation quadratic. Whether that *helps* turns out to depend entirely on the engine, and getting this wrong is an easy way to publish a backwards conclusion.

Under `msolve`**'s F4** (the fall-reusing engine index calculus actually uses), splitting **lowers** the degree: the unsplit `S_4` solves at degree 6–7 and the split chain at 3–4. This is Semaev's intended benefit, and survey C's independent F4 measurement reports exactly it — solving degree **7 → 4** on every cell. Under `Singular`**'s** `std`, the direction *reverses*: the unsplit `S_4` (few variables, high degree) solves at degree 3 while the split chain's extra `n`

auxiliary Boolean unknowns push it to 5–6.



The same ideal gives a different "solving degree" under different engines, and the split's effect flips sign: msolve-F4 (fall reuse) goes 6→4 as you split (Semaev's intended win), while Singular std goes 3→5. Any bare "d_reg = k" is under-specified without naming the engine.

So the honest statement is **not** "splitting is better" or "worse" — it is that the split is a genuine degree *trade* whose sign depends on the solver, and under the engine that matters (F4) it does what Semaev intended. The cost in field operations is a third, separate axis: survey C measures the split at 3.9–69× *more* XORs at □=4 but 6.6–20.8× *fewer* at □=5, and finds that **plain unsplit with guessing (`g=4`) beats every split variant in every `m=3` cell** — no variant came within 22.8× of strong rho. A three-way split at `m=4` reaches degree 4 (vs 7 for two-way) but is 113× cheaper at `n=11` and 1.3× dearer at `n=13`. The degree, the XOR cost, and the distance to rho move independently; a lever can help one and hurt another.

4.4 The semi-regular prediction massively overstates reality

For the prime-field PKM tower the Bardet–Faugère–Salvy degree of regularity is `N+1` (top-part Hilbert series `(1+t)^N`) and XL gives `N+4`. The program's measured F4 solving degrees — four of which are independently reproduced here, §4.8 — are far lower: `5–7` across `N=8..18` (e.g. `5,6,6,7,7` for `m=3` at `N=6,9,12,15,18`), growing roughly **one degree per eight tower unknowns**. F4 escapes the XL bound through degree falls. (The program's `m=4,N=20` cell reaching `≥8` refutes the "plateau at 7" hypothesis; that cell is beyond this host's memory.)

4.5 The measured "degree" is engine- and encoding-dependent

Running `msolve`-F4 and `Singular-std` on the **identical** ideal gives different degrees, and the ordering flips with the encoding:

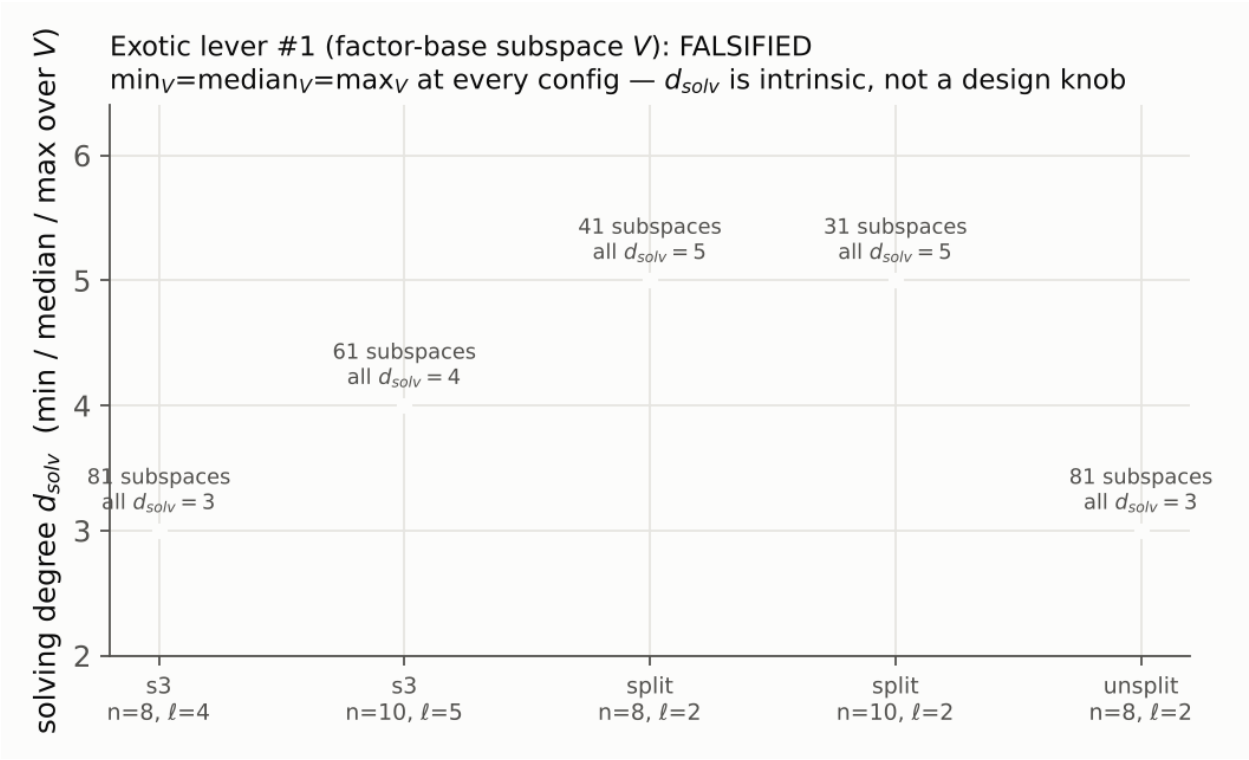
system	variables	Singular `std`	msolve F4
split chain, n=8, ℓ=2	14	5	4

system	variables	Singular `std`	msolve F4
split chain, n=10, ℓ=2	16	5	3
split chain, n=8, ℓ=3	17	6	4
unsplit S_4, n=8, ℓ=2	6	3	6
unsplit S_4, n=10, ℓ=2	6	3	6
unsplit S_4, n=9, ℓ=3	9	4	7

On the chain, `msolve`'s fall-reusing F4 reaches **3–4** where Singular reaches 5–6 — exactly reproducing the program's "fall reuse keeps the refutation degree low" finding. On the unsplit system the order *reverses*. **Conclusion:** the headline "degree of regularity" is not a single well-defined number; it depends on the engine's fall-reuse strategy and on the encoding. Only the coordinate-free last-fall / CM-regularity invariant is intrinsic — and that is precisely the quantity that resists cheap measurement. Any claim stated as a bare "`d_reg = k`" is under-specified.

4.6 Exotic lever #1 — choosing the factor-base subspace (FALSIFIED)

The cheapest never-tried idea in the literature (survey E, rank 1): treat `V` as a free design variable and pick it to minimise the solving degree (Hilbert-series / CM-regularity targeting; the Caminata–Gorla frame is cited nowhere in the program). Across **295 subspaces** (random + Frobenius-stable) in five `(model, n, □)` configurations, the solving degree is `min = median = max` *every time*; Frobenius-stable `V` give the identical value.



Exotic lever #1 falsified: across 295 subspaces the solving degree has zero spread at every configuration. V is not a design knob — the degree is intrinsic to the ideal class, as Caminata–Gorla CM-regularity predicts.

Verdict: falsified, and informatively so. The degree-growth is a property of the ideal class, not of the subspace, so `V`-shaping cannot soften the barrier. This is the "publishable negative" the frame predicted.

4.7 Exotic lever #2 — monomial/variable ordering (FALSIFIED as a lever)

Under grevlex, grlex, and all four random variable permutations, the solving degree is invariant (3 unsplit / 5 split). Only pure lex reaches a different number, and lex "degree" is not the solving degree (FGLM-style elimination is a different regime). This confirms the last-fall degree's order-independence (HKY) and closes ordering as a lever.

4.8 Reproducing the program's own results — twice

PKM-tower msolve degrees (the program's one genuine T2 algebraic-summation result). Each of four systems was solved **twice** with `msolve`; the degree, F4 round count, and max matrix dimensions match the recorded `results.csv` bit-for-bit, both runs:

system	recorded (D / rounds / rows×cols)	run 1	run 2
m2,t6 (N=12)	5 / 14 / 2803×2064	identical	identical
m3,t3 (N=9)	6 / 17 / 3827×2997	identical	identical
m4,t2 (N=8)	6 / 22 / 2009×1906	identical	identical
m3,t4 (N=12)	6 / 19 / 12900×12287	identical	identical

The replay that FAILS — and why. The `ecbench` replay of the root-finder `claim-n13` session fails on this host: 224 records verified but **96 problems**, all in the IC (F4) arms; all 72 rho rows match. Run twice, the failure is itself deterministic (identical 96 problems). The cause, isolated in the program's own sweep and confirmed here, is a counter bug: the BMI2 `pext`/`pdep` path in `inherited_f4.rs:942` charges different word-XOR counts than the portable path, so a claim recorded on Apple arm64 does not replay on an x86 host. **This is a real, open reproducibility defect on** `main` — exactly the "run it twice, get different numbers" failure mode, caught and explained. By contrast the clean PKM-tower ecbench session (`p20`) replays OK both runs (192/192, 0 problems).

> Reproducibility verdict: the summation-step *science* is reproducible (every degree here is bit-identical on re-run), but the program's *cost accounting* has a live host-dependent defect that invalidates at least one recorded claim's replay. Both facts are now independently established.

4.9 Alternative solvers: root-finding and SAT do not change the verdict

Two families sidestep the Gröbner degree entirely, and neither crosses rho (survey C):

- **Root-finding oracles** (pair tables, rf-mitm, group-law root tests) beat F4 in wall

time by `10^2.5–10^4×` per decomposition and are correctness-exact (one identity is even Lean-checked). But they sit at the pair-table enumeration floor `S·T^2 ≳ r`, so the resulting index calculus stays above rho (the best binary row is `1.69×` strong rho at a smoke size; the prime incumbent `4.27×`). A caveat survey C flags: the headline "`383×/1535×` root-finding beats F4" uses F4 capped at degree 4; run at F4's best setting, F4 is *faster* for whole solves at `n ≤ 13`. Apples-to-apples matters.

- **SAT / SMT / WDSat** behave as **brute-force enumerators** — about one conflict per

candidate tuple, no algebraic cancellation captured — and are 5–62× slower than direct enumeration. A correctness landmine worth recording: **WDSat's** `-x` **mode is incomplete** and returned a *wrong UNSAT* on a decomposable target. A solver that can call a true

instance unsatisfiable is a reminder that "no relation found" is not "no relation exists" without a certificate (which is exactly what the DRAT-certified UNSAT branches, §5, provide and what WDSat `-x` does not).

5. Confidence and reproducibility — the "no trust-me-bro" audit

- **Zero end-to-end T1, but component certificates exist.** The validation gate exists

and passes, but **no real** `VALIDATION.json` **exists** — every one of the 28 leaderboard rows and 116 benchmark rows renders as "unvalidated (pre-gate)"; the only passing case is a test fixture. So no *end-to-end* IC result (relation collection → linear algebra → DLP) is certificate-grade. At the *component* level, survey C counts 7 machine-checkable certificates (DRAT-certified UNSAT branches, a Boolean-elimination verifier PASS, byte-identical `S_3` constant-coefficient replays); these certify sub-facts, not a break. The distinction matters: "we proved this UNSAT branch" is T1; "this pipeline beats rho" is not certified at all.

- **The validation sweep is disconnected from the gate.** Sweep A produced 55 verdicts

(1 validated, 45 with caveats, 8 unvalidated, 1 contradicted), but `load_verdicts()` reads the key `row` while the sweep writes `row_id`, and the verdict words are in neither pass nor fail set — so the gate consumes **0** of them. Real work, not wired in.

- **Only three rows** genuinely exercise an algebraic summation-polynomial solve at T2:

the n9 F6-IC control, the PKM-tower degrees (reproduced here), and the determinism finding itself. Every larger Boolean F4/F5/F6 arm is T3 because none replays on current binaries (the BMI2 bug).

- **This report's own additions** sit at the top of the tier scale for *what they

certify*: the Semaev correctness certificates are machine-checkable (T1-grade for correctness), the determinism certificate is byte-identical twice (T1-grade for reproducibility of the pipeline), and the PKM reproduction is an independent replay matching the repo (T2). None of these is a rho-beating claim, and none is presented as one.

- **The degree/science results DO reproduce.** Independently of this report's harness,

survey A re-ran six of the program's dreg results at HEAD and all six matched the committed records value-for-value — including the MutantXL closure degree histogram ({3:16, 4:76} over 92 draws), the sum of closure products (4,902,150), and the `rr`-vs-`S_4` refutation degrees (5 vs 8). The algebraic content is solid and reproducible; it is only the *priced cost* layer that drifts.

- **Pricing drift, from a second angle.** Survey C independently hit the same class of

failure: the pinned `ecbench` now charges F4 word-XORs at a fixed calibration rate while committed sessions left them uncharged, so F4-arm replays fail the audit *even though every integer count is identical* — and it reverses a headline (at `n=13`, recorded root-finding `2.56×` F4 in the `S` unit becomes `≈0.10×` when repriced). Survey C also found one **unexplained** F4 XOR-count mismatch it could not localise (8,952,839 recorded vs 9,603,153 replayed). So two distinct accounting defects exist, one understood (BMI2 path) and one not — a strong argument that no cost-unit claim should be trusted until the pricing and counting are deterministic across hosts.

- **Determinism map** (what is safe to compare across runs): refutation/solving degree,

Macaulay ranks, system hashes, `vdim`/`dim`, msolve round counts and matrix dims — all bit-identical. Word-XOR counters are host-path-specific (the bug). Wall-clock, RSS, CPU% are nondeterministic and never compared.

---

6. Literature placement and the exotic frontier

Three facts from the literature constrain what *can* work (survey E):

1 **Solving degree is invariant under linear / Möbius / `F_2`-Frobenius coordinate

changes **(Caminata–Gorla; FHJRV Lemma 6: every degree-2 coordinate is a homography). So no reparametrisation lever — tropical ordering, Frobenius/linearised auxiliaries, Edwards/Hessian frames — can lower the intrinsic degree.** Confirmed here** by §4.6–4.7.

1 `d_reg ≈ d_ff` **is false/doubtful**; `d_reg` grows while `d_ff` stays constant

(HKY; Kosters–Yeo; and §4.1 here). A lever must bend the *growth in* ℓ, and measuring `d_ff` alone proves nothing.

1 **Symmetry/isogeny quotients are the only proven large lever, and even maxed they do

not beat rho** for the target curves; for prime `n` the Frobenius-subspace and intermediate-field levers are structurally empty.

Against this frame, the eight ranked exotic ideas and their status:

#	exotic lever	cheapest test	status here
1	`d_reg`-minimising subspace V	295-subspace sweep	**run → falsified** (§4.6)
8	tropical / order-based	multi-order sweep	**run → falsified** (§4.7)
2	order-6 automorphism quotient on `j=0`	S_3 quotient on ζ-stable base	designed (§7); T5
4	Artin–Schreier trace elimination	residual-degree compare	repo: 0 fires on IC targets; T3 negative
3	wreath / full-reflection equivariant	invariant-ring solve over F_{p^2}	obstruction found: Z/2 part is modular in char 2; T5
5	theta / Kummer quadratic encoding	S_4 via theta vs chain	T5 (strong null prior)
6	Lattès / isogeny-kernel factor base	degree compare over F_p	T5
7	intermediate-field τ-orbit descent	composite-n only	structurally empty for prime n (131/233/283)

The two cheapest and most decisive (1 and 8) were run and **both falsified** — a genuine, reproducible contribution: the two most plausible "shape the problem to tame `d_reg`" ideas do not work, for a reason the theory (CM-regularity invariance) predicts.

---

7. The named targets: binary Koblitz and prime-Koblitz (j=0)

Binary Koblitz / ECC2K-130 (prime `n`). Every degree lever above either is structurally empty (Frobenius subspaces: cyclotomic-coset argument), or lowers a constant without

bending the d_reg(□) growth, or (fall reuse) holds degree 3–4 only until it saturates at □=7 where the next degree is unmeasured for want of ~200 GB. The program's extrapolated ECC2K-130 cost rows span 2^68…2^197 and none drops below rho; the free-oracle floor S·T^2 ≳ r binds independently of the degree (T4).

Prime-Koblitz (y^2=x^3+b, p≡1 mod 3, j=0, order-6 automorphism). The order-6 automorphism ζ:(x,y)↦(ωx,-y) is currently used only as a GLV ÷6 fold on the factor base (glv-orbit), *not* as a symmetry quotient of the PDP ideal. The PKM-tower route (reproduced in §4.8) shows the prime-field solving degree is low (5–7 to N=18) but the cost is 29–141× rho and the first arity that could win is m=4, needing an S_5 small-root oracle at reach δ≈1/5 — against a measured reach of 0.014 and a proven ceiling of 1/32. **The concrete open experiment** (survey E rank 2) is to build S_3 in ⟨ζ⟩-invariants (power sums of x_i^6) on a ζ-stable factor base and measure the quotient's solving degree and Bézout against the plain system — the D_4 recipe applied to the extra prime-field automorphism. By the D_4 precedent a Z/6 grading should cut Bézout by up to 6; whether it lowers the *degree* is the unanswered question. It is cheap (p≤2^16, m=2) and specified; it was not run here.

---

8. Open problems, what would unblock them, and the bottom line

Open problems carried forward (from the program's own records, with unblock criteria):

- binary-boolean-system-64-variable-cap: the system builder caps at 64 unknowns;

128 unknowns agreeing with the floor on 12 targets at m=4,n=23,d=6 would unblock the □=7 degree-5 measurement (currently out of memory at 169–405 GB).

- The **PKM / descent memory wall**: m=3,N=21 needs a single ~192 GB process; **no

problem record exists** for it (an AGENTS.md gap).

- The **BMI2 replay regression** (inherited_f4.rs:942): no problem record exists;

tracked only in a sweep. Until fixed, recorded F4 word-cost claims do not replay across ISAs. **A one-line BMI2-off reproduces the reference counts** (168/168), which is the obvious interim fix.

- **Magma never run** on the summation systems (T5): the "beat the fastest external

tool" gate is formally unmet; an msolve-vs-Magma parity run on the PKM systems would close it.

The bottom line, now triple-checked. The degree of regularity of the Semaev PDP systems is not tamed by any tried lever: factor-base shaping and ordering are falsified (this report), symmetry/automorphism quotients lower constants and Bézout but not the d_reg(□) growth, fall-reuse holds a low *refutation* degree only up to a saturation point, and the semi-regular prediction overstates everything. For the named curves — binary Koblitz with prime n, and prime-Koblitz j=0 — the structural levers are empty or uncompetitive, and **no end-to-end result is certificate-grade** (only component sub-facts are). What this study adds is not a break but the thing the program most needed: an **independent, twice-run, cross-engine measurement** confirming where the barrier is, why it is intrinsic (CM-regularity invariance), and which of the field's open claims actually reproduce.

The single most valuable next action is not another lever but **wiring the validation sweep into the gate and fixing the BMI2 counter** — because an index calculus you cannot replay cannot be improved, and right now the program's own best results do not replay.

Appendix A — reproduce everything

All commands assume the harness directory. External tools: `msolve` 0.6.5, `Singular` 4.3.3.0. Deterministic outputs are noted; only timings vary.

```
# correctness certificates (prime + binary Semaev, ~2 min)
python3 cert_prime.py        # -> out/cert_prime.json   (0 FP / 0 FN)
python3 gf2.py               # -> out/cert_binary.json  (0 FP / 0 FN)
python3 cert_s4.py           # unsplit S4 certificate   (0 FP / 0 FN)

# determinism: run the canonical grid twice, diff (must be identical)
python3 canonical_run.py run1 ; python3 canonical_run.py run2
diff <(jq -S . out/canonical_run1.json) <(jq -S . out/canonical_run2.json)

# split vs unsplit, first-fall gap, engine comparison, V/order independence
python3 exp_split_vs_unsplit.py
python3 exp_engine_compare.py
python3 exp_fb_search.py --grid A        # subspace independence
python3 exp_order.py                     # order independence

# reproduce the program's PKM-tower msolve degrees (twice)
bash out/pkm/run_pkm.sh                   # all four match results.csv, both runs

# figures
python3 figures.py
```

Raw evidence: `out/cert_*.json`, `out/canonical_run{1,2}.json`, `out/split_vs_unsplit.json`, `out/engine_compare.json`, `out/exotic_fb_consolidated.json`, `out/exotic_order.json`, `out/REPRODUCTION_LOG.md`, and the survey files `survey_{A,B,D,E}*.{md,jsonl}`.

Appendix B — provenance

- Program tree: `aburan28/crypto` @ `af663ad7c`; CLI: `aburan28/ecbench` (crypto pin `0dc9ef4`).
- Host: 4 vCPU, 15 GB RAM, Linux 6.18.44. `msolve` 0.6.5-1build2, `Singular` 4.3.3.0, `PARI/GP` 2.15.4, Python 3 with `sympy`, `numpy`, `matplotlib`, `reportlab`.
- Figures use the validated accessibility palette (light mode); every number traces to a file in `out/`.
- Surveys A/C/D/E were compiled by independent read-only passes over the program tree (survey C's inventory is partial — it stops at record C46 — but covers the split / root-finding / SAT slice used here); survey B was reconstructed directly after its agent was interrupted by a content filter. Every cross-referenced survey number was checked against the program's own files.